

Data Warehousing and Data Mining

Rapid-Revision One-Liners

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2026 EDITION
VERSION 1.0

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Data Warehousing and Data Mining - One-Liners

Data Warehouse - Definition and Characteristics

1. A data warehouse has **4** defining characteristics: **subject-oriented**, **integrated**, **time-variant** and **non-volatile**.
2. The **1** person credited as "Father of Data Warehousing" is **Bill Inmon**, who gave the definition in **1990**.
3. There are **2** allowed operations on a non-volatile warehouse: **initial loading** and **read/query access**.
4. A warehouse typically retains **5-10 years** of **historical (time-variant)** data versus an operational DB's **current** data.
5. The "**subject-oriented**" trait organizes data around **subjects** (customer, product, sales), not around **applications**.
6. The "**integrated**" trait demands consistency in **4** things: **naming**, **encoding**, **units** and **formats** across sources.
7. The mnemonic for the **4** warehouse keywords is "**SITN**": **Subject-oriented, Integrated, Time-variant, Non-volatile**.
8. The core purpose of a data warehouse is **1** thing: supporting management **decision-making** (DSS).
9. A warehouse provides exactly **1** organizational benefit called the "**single version of the truth**".

OLTP vs OLAP

10. The **2** complementary processing styles are **OLTP** (transaction processing) and **OLAP** (analytical processing).
11. OLTP is **application-oriented** while OLAP is **subject-oriented** - the **2** opposite design orientations.
12. OLTP holds **current detailed** data; OLAP holds **historical summarized** data - a **2-way** data contrast.
13. OLTP databases are usually **normalized (3NF)** while OLAP uses **denormalized** star/snowflake - **2** modeling styles.
14. An OLTP query touches about **tens** of records; an OLAP query may scan **millions** - the access-volume gap.
15. OLTP serves **thousands** of clerical users; OLAP serves **hundreds** of analysts/managers.
16. OLTP size is typically **MB-GB**; an OLAP warehouse is **GB-TB-PB** in size.
17. The performance metric for OLTP is **transactions/second**; for OLAP it is **query throughput / response time**.
18. OLTP supports **3** write operations (**INSERT, UPDATE, DELETE**) while OLAP is **mostly read-only**.

Data Warehouse Architecture

19. The standard data-warehouse architecture has **3** tiers: **bottom (DB server)**, **middle (OLAP server)** and **top (front-end tools)**.
20. The **bottom tier** is almost always **1** kind of engine - a **relational DBMS** - fed by **ETL** back-end tools.
21. The **middle tier** is the **OLAP server**, implemented in **3** ways: **ROLAP, MOLAP** or **HOLAP**.
22. The **top tier** hosts **4** front-end tool classes: **query**, **reporting**, **analysis** and **data mining**.
23. A data cube's **2** building blocks are **dimensions** (perspectives) and **measures/facts** (numbers).
24. The lowest-detail cuboid is the **base cuboid** and the fully aggregated one is the **apex cuboid** - **2** extremes of the cube lattice.

Warehouse Strategy, Planning and Management

25. There are **2** classic warehouse-build strategies: **Top-Down (Inmon)** and **Bottom-Up (Kimball)**.
26. **Inmon's top-down** builds the **enterprise warehouse first** then marts, using a **normalized (3NF)** model.
27. **Kimball's bottom-up** builds **data marts first** then integrates, using a **dimensional (star)** model.
28. The bottom-up approach wins on **2** counts: **lower cost** and **faster delivery**, but risks integration later.
29. Warehouse planning follows broadly **1** flow: **requirements -> modeling -> source mapping -> ETL -> build -> deploy -> maintain**.
30. Warehouse support processes include at least **6**: **scheduling**, **backup**, **security**, **monitoring**, **tuning** and **capacity planning**.

Warehouse Infrastructure

31. The **3** parallel hardware architectures are **SMP, MPP** and **NUMA**.
32. **SMP** shares **2** resources - **memory and disk** (shared-everything, tightly coupled, limited scale).
33. **MPP** is a **shared-nothing** design where each node owns its **own memory and disk** - the most scalable.

- 34. **NUMA** uses **shared memory** but with **non-uniform** access where local memory is **faster** than remote.
- 35. The **2** parallel performance terms are **speed-up** (faster on fixed job) and **scale-up** (bigger job, same time).
- 36. Query parallelism comes in **2** forms: **inter-query** (many queries at once) and **intra-query** (one query split).
- 37. A distributed DBMS supports **3** transparency types: **location**, **fragmentation** and **replication** transparency.
- 38. Data fragmentation has **3** forms: **horizontal** (rows), **vertical** (columns) and **mixed/hybrid**.
- 39. Warehouse storage redundancy is commonly provided by **RAID** arrays and scalable **SAN** storage - **2** I/O technologies.

Schema Design

- 40. The **3** warehouse schema types are **Star**, **Snowflake** and **Fact Constellation (Galaxy)**.
- 41. A **star schema** has **1** central **fact table** surrounded by **denormalized dimension** tables.
- 42. A **snowflake schema** differs by **1** change - its **dimension tables are normalized** into sub-tables.
- 43. A **fact-constellation/galaxy** schema has **multiple fact tables** that **share** common dimension tables.
- 44. Among the **3** schemas, the **star** has the **fewest joins** and **fastest** queries but **most** redundancy.
- 45. The **snowflake** trades **2** things - **less storage/redundancy** for **more joins** (slower queries).
- 46. A fact table stores **2** things: numeric **measures** and **foreign keys** to dimensions; it is **deep and narrow**.
- 47. A dimension table stores **descriptive textual attributes** and is **wide with fewer rows**.
- 48. Facts come in **3** additivity types: **additive**, **semi-additive** and **non-additive**.
- 49. A **semi-additive** fact (e.g. account **balance**) can be summed over **all dimensions except time**.
- 50. The detail level of one fact-table row is called its **grain** - exactly **1** key design decision.

ETL - Extract, Transform, Load

- 51. **ETL** stands for **3** stages in order: **Extract**, **Transform** and **Load**.
- 52. The **Extract** stage reads from **heterogeneous source** systems in **2** modes: **full** and **incremental**.
- 53. The **Transform** stage is the **heaviest** and does **5+** tasks: **clean**, **standardize**, **deduplicate**, **derive** and **aggregate**.
- 54. The **Load** stage writes data in **3** ways: **initial load**, **incremental load** and **full refresh**.
- 55. The modern variant **ELT** reorders to **Load-then-Transform**, doing transformation **inside** the target engine.
- 56. Common ETL tools include **4+**: **Informatica**, **SSIS**, **DataStage** and **Talend**.

Metadata and Data Marts

- 57. Metadata means **1** thing - "**data about data**" - and lives in the **metadata repository**.
- 58. There are **3** metadata types: **technical**, **business** and **operational** metadata.
- 59. **Technical** metadata describes **source schemas and ETL mappings**; **business** metadata describes **terms and ownership**.
- 60. **Operational** metadata tracks **3** items: **data lineage**, **currency** and **ETL job logs**.
- 61. A data mart is a **subset** of the warehouse focused on **1** subject or department.
- 62. Data marts come in **2** types: **dependent** (fed from the central DW) and **independent** (built from sources directly).
- 63. The **dependent** mart sources **from the warehouse**; the **independent** mart **bypasses** the warehouse.

Data Mining - Introduction

- 64. Data mining is also called **KDD - Knowledge Discovery** from Data - extracting **non-trivial, useful** patterns.
- 65. A useful mined pattern must satisfy **4** properties: **valid**, **novel**, **useful** and **understandable**.
- 66. The motivation phrase for data mining is **1** slogan: "**data rich but information poor**".
- 67. Data mining sits inside KDD as exactly **1** step among the **7** total steps.

KDD Process

- 68. The KDD process has **7** steps: **cleaning**, **integration**, **selection**, **transformation**, **mining**, **evaluation** and **presentation**.
- 69. The first **2** KDD steps (**cleaning**, **integration**) plus selection/transformation together form **data preprocessing**.

- 70. The core pattern-finding KDD step is step **5**, the actual **data mining** step.
- 71. The last **2** KDD steps are **pattern evaluation** and **knowledge presentation (visualization)**.

Data Mining Functionalities

- 72. Mining tasks split into **2** broad classes: **descriptive** (general properties) and **predictive** (inference).
- 73. **Class/concept description** has **2** forms: **characterization** (one class) and **discrimination** (compare classes).
- 74. The **6** main data-mining functionalities are **description, association, classification, clustering, outlier analysis** and **evolution analysis**.
- 75. **Classification** predicts **categorical** labels while **regression/prediction** predicts **continuous** values - **2** predictive tasks.
- 76. **Outlier analysis** finds **deviating** objects and is the basis for **1** key application - **fraud detection**.
- 77. **Evolution analysis** models **trends over time**, i.e. **time-series** patterns.

Data Mining Architecture

- 78. The data-mining architecture has about **6** components, the core being the **data mining engine**.
- 79. The **knowledge base** supplies **domain knowledge** used to **guide and evaluate** the search.
- 80. The **pattern evaluation module** filters results using **interestingness measures**.
- 81. Data sources for mining include **3** kinds: **databases, data warehouses** and the **WWW**.

Data Mining vs Warehousing vs OLAP

- 82. OLAP is **verification/query-driven** while data mining is **discovery-driven** - the **2** key analytical mindsets.
- 83. OLAP answers "**what happened**"; data mining answers "**why and what will happen**" - a **2-way** contrast.

Data Preprocessing - Overview

- 84. Real-world data has **3** quality problems: it is **incomplete, noisy** and **inconsistent**.
- 85. The **4-5** forms of preprocessing are **cleaning, integration, transformation, reduction** and **discretization**.
- 86. The preprocessing mnemonic "**CITRD**" lists **5**: **Cleaning, Integration, Transformation, Reduction, Discretization**.
- 87. The data-quality slogan that motivates preprocessing is **1** phrase: "**garbage in, garbage out**".

Data Cleaning - Missing Values and Noise

- 88. There are about **6** ways to handle missing values, the **best** being "**use the most probable value**".
- 89. Filling a missing value with the **attribute mean for the same class** is smarter than the **global mean** - **2** of the methods.
- 90. **Noisy data** is defined as **1** thing: a **random error or variance** in a measured variable.
- 91. The **3** main noise-smoothing techniques are **binning, regression** and **clustering**.
- 92. **Binning** first does **1** prerequisite - **sorting** - then partitions into **equal-width** or **equal-frequency** bins.
- 93. After binning, smoothing uses **3** options: **bin means, bin medians** or **bin boundaries**.
- 94. **Clustering** smooths noise by treating points **outside clusters** as **outliers** to remove.
- 95. A **4th** noise method combines **computer and human inspection** for suspicious values.
- 96. **Inconsistent data** is fixed using **2** aids: **external references** and **domain knowledge / functional dependencies**.

Data Integration and Transformation

- 97. The integration **entity identification problem** matches **equivalent attributes** like **cust_id** and **customer_no**.
- 98. **Redundancy** between attributes is detected by **correlation analysis** using **2** tests: **chi-square** (nominal) and **correlation coefficient** (numeric).
- 99. Data-integration conflicts include **2** common kinds: **tuple duplication** and **data value conflicts** (units/scales).
- 100. Data transformation includes **5** operations: **smoothing, aggregation, generalization, attribute construction** and **normalization**.
- 101. There are **3** normalization methods: **min-max, z-score** and **decimal scaling**.

102. **Min-max** normalization maps a value to $[new_min, new_max]$ via $v' = (v - min) / (max - min) * (new_max - new_min) + new_min$.
103. **Z-score** normalization uses $v' = (v - mean) / std_dev$ and is preferred when **min/max are unknown** or outliers exist.
104. **Decimal scaling** uses $v' = v / 10^j$ where j is the smallest integer making $max(|v'|) < 1$.

Data Reduction and Discretization

105. Data reduction has **5** techniques: **cube aggregation**, **dimensionality reduction**, **numerosity reduction**, **compression** and **discretization**.
106. **Dimensionality reduction** removes **attributes (columns)** using **3** tools: **wavelets (DWT)**, **PCA** and **attribute subset selection**.
107. **PCA** is also called the **Karhunen-Loeve** method and is a **lossy** reduction technique.
108. **Numerosity reduction** reduces **rows/values** in **2** styles: **parametric** (regression, log-linear) and **non-parametric** (histograms, clustering, sampling).
109. Sampling for numerosity reduction includes types like **SRSWOR**, **SRSWR** and **stratified** sampling.
110. Data compression has **2** kinds: **lossless** (fully recoverable) and **lossy** (approximate, e.g. DWT/PCA).
111. **Discretization** converts a **continuous** attribute into **intervals**, and is **supervised** or **unsupervised**.
112. Discretization can be **top-down (splitting)** like **entropy-based**, or **bottom-up (merging)** like **ChiMerge** - **2** directions.
113. A **concept hierarchy** maps low-to-high concepts, e.g. **street < city < state < country** - **4** levels.
114. The intuitive numeric-partitioning rule for concept hierarchies is the **3-4-5** rule.

Data Visualization and OLAP Operations

115. The **3** warehouse essentials emphasized in visualization are **aggregation**, **historical information** and **query facility**.
116. **OLAP** stands for **Online Analytical Processing** and was characterized by **E. F. Codd's 12 rules** in **1993**.
117. The alternative OLAP test is **FASMI = Fast Analysis of Shared Multidimensional Information**.
118. There are **5** core OLAP operations: **roll-up**, **drill-down**, **slice**, **dice** and **pivot**.
119. **Roll-up (drill-up)** does **1** thing - **aggregates** by climbing a hierarchy or reducing a dimension.
120. **Drill-down** is the reverse, moving to **finer detail** or **adding a dimension**.
121. **Slice** selects on **1** dimension producing a sub-cube; **dice** selects on **2 or more** dimensions.
122. **Pivot (rotate)** simply **reorients** the cube axes - e.g. swapping **rows and columns**.

OLAP Servers - ROLAP, MOLAP, HOLAP

123. There are **3** OLAP server types: **ROLAP** (relational), **MOLAP** (multidimensional) and **HOLAP** (hybrid).
124. **ROLAP** stores data in **relational tables** and computes aggregates **on the fly via SQL**, scaling to **very large** data.
125. **MOLAP** stores data in **pre-computed multidimensional cubes**, giving the **fastest** queries.
126. **MOLAP's** main weakness is **1** issue - the **sparse-matrix / cube explosion** problem.
127. **HOLAP** keeps **detailed data in relational** storage and **aggregations in MOLAP** cubes - the best-of-both hybrid.
128. The mnemonic links each by initial: **ROLAP=Relational**, **MOLAP=Multidimensional**, **HOLAP=Hybrid** - **3** mappings.

Association Rule Mining

129. An association rule **A=>B** is judged by **3** measures: **support**, **confidence** and **lift**.
130. **Support(A=>B)** equals **1** probability: **P(A U B)** - the fraction of transactions with both items.
131. **Confidence(A=>B)** equals **P(B|A) = Support(A U B)/Support(A)** - the rule's reliability.
132. **Lift(A=>B)** equals **Support(A U B)/(Support(A)*Support(B))** and has **3** readings around the value **1**.
133. **Lift = 1** means **independent**, **Lift > 1** means **positively** correlated, **Lift < 1** means **negatively** correlated.
134. A **strong rule** must pass **2** thresholds simultaneously: **min_support** and **min_confidence**.
135. A **frequent itemset** is **1** condition: it has **support >= min_support**.

136. The classic association use-case is **1** scenario - **market-basket analysis**.

Apriori and FP-Growth

137. The **Apriori** algorithm was given by **Agrawal and Srikant in 1994** and is a **level-wise** method.
138. The **Apriori property** is **anti-monotone**: **every subset of a frequent itemset is frequent** (and supersets of infrequent are infrequent).
139. Apriori repeats **2** steps per level: **join** (build candidates C_k) and **prune** (drop infrequent-subset candidates).
140. Apriori's **2** main drawbacks are **multiple database scans** and a **huge candidate set**.
141. **FP-Growth** (Han, Pei, Yin, **2000**) needs only **2** database scans and uses **no candidate generation**.
142. FP-Growth builds **1** compact structure - the **FP-tree** - and mines via **conditional pattern bases**.
143. The headline difference is **1**: Apriori **generates candidates**, FP-Growth **does not**.

Classification

144. Classification is **supervised** and works in **2** steps: **model construction (training)** and **model usage (testing)**.
145. The **3** famous decision-tree algorithms are **ID3**, **C4.5** and **CART**.
146. **ID3** splits using **Information Gain (entropy)**; **C4.5** uses **Gain Ratio**; **CART** uses the **Gini Index**.
147. Dataset impurity is measured by **entropy** = $-\sum p_i \log_2(p_i)$ - **1** core formula.
148. **Information Gain** = $\text{Info}(D) - \text{Info}_A(D)$; ID3 picks the attribute with the **highest** gain.
149. The **Gini Index** = $1 - \sum (p_i)^2$; CART picks the **lowest** Gini split and builds **binary** trees.
150. **ID3** has **1** known bias: it favors attributes with **many distinct values**, which **Gain Ratio** corrects.
151. Decision-tree overfitting is controlled by **pruning** in **2** styles: **pre-pruning** and **post-pruning**.
152. **Bayesian classification** rests on **Bayes' theorem**: $P(H|X) = P(X|H)P(H)/P(X)$.
153. **Naive Bayes** makes **1** simplifying assumption - **class-conditional independence** of attributes.
154. In Bayes' theorem the **4** terms are **posterior $P(H|X)$** , **prior $P(H)$** , **likelihood $P(X|H)$** and **evidence $P(X)$** .
155. **k-NN** is a **lazy / instance-based** learner that classifies by the **majority vote of k neighbours**.
156. k-NN usually measures closeness with **1** metric - the **Euclidean distance**.
157. The **2** learner styles are **lazy** (k-NN, works at query time) and **eager** (decision tree/Naive Bayes, builds model at training time).
158. **Neural networks** train via **backpropagation** but are criticized as **1** thing - a **"black box"**.

Clustering

159. Clustering is **unsupervised** with **1** goal: **high intra-cluster** and **low inter-cluster** similarity.
160. The major clustering families are **4**: **partitioning**, **hierarchical**, **density-based** and **grid-based**.
161. The **2** partitioning methods are **k-Means** (centroid/mean) and **k-Medoids/PAM** (representative object).
162. **k-Means** complexity is $O(nkt)$ for **n** points, **k** clusters and **t** iterations.
163. **k-Means** has **2** weaknesses: it is **sensitive to outliers/noise** and needs **k** specified in advance.
164. **k-Medoids** is **more robust to outliers** than k-means but costs about $O(k(n-k)^2)$ per iteration.
165. Hierarchical clustering has **2** directions: **agglomerative (AGNES, bottom-up)** and **divisive (DIANA, top-down)**.
166. Hierarchical results are visualized with **1** diagram - the **dendrogram**.
167. **DBSCAN** is density-based and uses **2** parameters: **Eps** (radius) and **MinPts** (minimum points).
168. **DBSCAN** has **2** advantages: it finds **arbitrary-shaped** clusters and labels **noise/outliers**, without needing **k**.
169. The decisive contrast is **1**: **classification is supervised** (labeled) while **clustering is unsupervised** (unlabeled).

Prediction / Regression

170. **Regression** predicts a **continuous numeric** value, unlike classification's **discrete class**.
171. **Linear regression** fits **1** equation - $y = w \cdot x + b$ - by least squares.
172. Regression types include **3**: **simple linear**, **multiple linear** and **non-linear** regression.

Warehouse Management and Advanced Mining

- 173. Warehouse management covers **5** areas: **mining interface, security, backup/recovery, tuning** and **testing**.
- 174. Warehouse security uses **4** controls: **authentication, authorization, encryption** and **auditing**.
- 175. Tuning a warehouse relies on **4+** techniques: **indexing, partitioning, materialized views** and **aggregate pre-computation**.
- 176. **Bitmap indexes** suit **low-cardinality** columns while **B-tree indexes** suit **high-cardinality** keys - **2** index choices.
- 177. Backup strategies use **2** types - **full** and **incremental** - plus **checkpoints** for recovery.
- 178. The **3** warehouse application classes are **information processing, analytical processing (OLAP)** and **data mining**.

Web, Spatial, Temporal and Text Mining

- 179. **Web mining** has **3** categories: **content mining, structure mining** and **usage mining**.
- 180. **Web structure mining** analyzes **hyperlinks** using **2** algorithms: **PageRank** and **HITS (hubs & authorities)**.
- 181. **Web usage mining** mines **1** source - **server logs / clickstream** - to find **user access patterns**.
- 182. **Spatial mining** works on **GIS/geographic** data, finding **spatial association rules** and **co-location** patterns.
- 183. **Temporal mining** handles **time-stamped** data via **time-series analysis** and **sequential-pattern** mining.
- 184. **Text mining** extracts knowledge from **unstructured text** using **2** fields: **Information Retrieval** and **NLP**.
- 185. **Multimedia mining** targets **3** media types: **images, audio** and **video**.

Final High-Yield Recap

- 186. Remember the **4** Inmon keywords and the **3** schema types as the **2** most-asked warehouse facts.
- 187. The **5** OLAP operations and **3** OLAP servers (ROLAP/MOLAP/HOLAP) are near-certain MCQ topics.
- 188. Memorize **3** association formulas (**support, confidence, lift**) and the **2** algorithms (**Apriori, FP-Growth**).
- 189. Tie each tree to its measure: **ID3-InfoGain, C4.5-GainRatio, CART-Gini** - **3** fixed pairings.
- 190. Lock the **1** biggest contrast for the exam: **classification = supervised, clustering = unsupervised**.